

KAPSABET



SCHOOL

HIGH

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KAPSABET

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KB/ADM. NO. 12458

18/10/2022

Our Ref:.....

Date:.....

Dear Admissions Committee,

RE: RECOMMENDATION LETTER FOR OTIENO ROBERT ADMISSION NUMBER 12458

Most students nowadays lack genuine passion in their approach to learning Physics. 'K constant' is the term I peg such students. Robert, however, was not a 'K constant'. I regard him as one of the exceptional physics students of his class and an outstanding scientist in the Science club as he continuously demonstrated a good grasp of content beyond the typical syllabus. For instance, he used the concept of entanglement in quantum physics to explain how light travels from the source, to the object then to the eye, elaborating how all three exist in relation to each other. This ability to learn beyond curriculum made him an invaluable asset in my class.

Robert was an amalgam of high intellectual curiosity. His assorted and diversified interests always made him crave for more knowledge. It is in this that he utilized every resource in his environment to heighten his depth of knowledge. For example, he wasn't scared of bulky Physics textbooks that expounded on topics beyond our institutional level. Most of the time, the formulas were bombastic and he brought them onto me to help him understand them. I assisted him with the ones I could. He, however, never despaired on the ones that were beyond my level. He went to the computer and through various open coursewares posted by various institutions on youtube, he would try to understand the formulas. My colleagues and I grew fond of his presence and persistence. It is his curiosity that I believe gave him a superior intellectual edge in Physics. I was pleased when he participated in our school's yearly Physics Contest as a junior, a contest previously dominated by seniors. He wrestled with seniors for the top spot as the best student, and he would occasionally miss by just a point. Despite missing severally, this was an achievement as a junior. When he became a senior, he undoubtedly earned his position as the top student in Physics. Among his classmates, he would maintain his streak as the Physics subject champion in every exam. When the national exams were done, Robert emerged as the overall top student nationally in Physics with a score of 97. This feat made me proud to have been his Physics instructor. He managed to reaffirm his spot as the Physics subject champion, and because he scored 97, he won the Dean of Science Award in Physics. Robert was one of our most dedicated science students. Therefore, when I nominate him, it is my belief that he will perfectly fit into your STEM program.

During Physics practicals, it came to my knowledge that Robert possessed rigorous analytical skills. Using the first-hand knowledge I had given them in class, he would usually understand what an experiment was aiming at and establish relationships between and among investigated experimental variables. I once administered an experiment on Hooke's law of elasticity to determine the effect of force on elastic materials. The concept demanded an airtight analysis of the experimental variables which was force, the invariable which was elasticity and the apparatus set before them. I expected the class to draw a rising curve, insinuating that deformity on a material is directly proportional to the force applied on the material. The students however got a straight line instead of a curve. It was Robert who sought after me and told me that the weights used to apply the force were all equal therefore the deformity effect was the same creating a straight line instead of a flat curve. I was elated and told Robert to announce to everybody to approach the lab technician and request for varying weights. This makes me believe he will be a leader in academia who can thrive in experimentation-based research teams.

Robert's grasp of Physics' concepts was evident beyond the classroom. At the Kenya Science and Engineering Fair, it came to my realization that his hands-on problem-solving skills were outstanding. Using the idea of heat produced by an electric current through a conductor, Robert designed concepts of his 'anti-locust' drones right up from the ground, aiming at addressing the nation's locust saga. It was a drone-like craft that had electric wire mesh, which would fly in the opposite direction of swarms of locusts and when locusts came in contact with the mesh, they would be roasted as they dropped on the ground due to the heat produced by electric current through the mesh. Farmers would later harvest the roasted ones for human consumption. He proved the depth of his ingenuity was unmatched and despite other students despairing and presenting their error-stricken designs and models to the fairs, Robert persevered through the pressure to make sure his final product was distinctive. During interactions with him as his project instructor, I learnt Robert's perseverance throughout the project made him understand the harsh repetitive routines that come with being a scientist or an engineer. Through his 'anti-locust' drones, he partnered with Kenya Flying Labs, an organization that seeks to educate young Kenyan prospects on the entrepreneurial aspect of drone technology. With the resources and knowledge your institution will offer him, I am certain Robert will test his hands-on problem solving skills at different avenues for the betterment of society.

Robert was a humble altruistic student. He always volunteered to tutor his classmates in challenging physics topics. He took Benjamin Cheruiyot, a student who was consistently below average in Physics, and kept him under his tutelage. During lessons, Robert would switch his position to sit next to Benjamin so that when I gave the class time for discussions, Robert would revise with Benjamin the concepts I had taught while showing him how to answer related questions. Robert would also go out of his way to gather revision materials that would help Benjamin grasp concepts easily. In laboratory work sessions, Robert let Benjamin set the apparatus and work the measurements, preferring to let him explore, follow the procedures and only follow up when his set up had developed hiccups. When the national examinations were done, Benjamin scored an A- in Physics, an achievement given his previous academic records. I can't wait for Robert to recover his selfless nature at your institution. I will miss Robert's presence in class and in the laboratory. Given the resources, he will be granted and faculty he will interact with, I have no doubt he will shine at your institution.

Yours sincerely,

Mr. Isaac Choge,

HEAD OF SUBJECT- PHYSICS.

